I^{2}SB is a warm-started diffusion

ImMAP project notes

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Claim. With a regressor that knows $\boldsymbol{x}_1$, I^{2}SB [1] is a variance-exploding (VE) diffusion on the conditional prior $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)$, written in different coordinates. It (i) regresses $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$, (ii) re-noises it to a finite noise level, and (iii) runs the standard VE ancestral sampler [2] down to zero noise. The bridge only changes how each state is presented to the network.

One change of coordinates

The Brownian bridge has marginals $\boldsymbol{x}_t = (1-t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\,\boldsymbol{\varepsilon}$, $\sigma_{\mathrm{sb}}(t)^2 = Vt(1-t)$, $V = 4\tau^2$. Remove the $\boldsymbol{x}_1$ part and rescale:

$$\boldsymbol{u} \coloneqq \frac{\boldsymbol{x}_t - t\,\boldsymbol{x}_1}{1-t} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}, \qquad \sigma^2 = \frac{Vt}{1-t}, \qquad t = \frac{\sigma^2}{\sigma^2 + V}. \tag{1}$$

So bridge time *is* a noise level, with $t \in [0,1) \leftrightarrow \sigma \in [0,\infty)$, and $\boldsymbol{u}$ is a VE-diffused $\boldsymbol{x}_0$. Conversely,

$$\boldsymbol{x}_t = \boldsymbol{x}_1 + c(\sigma)\,(\boldsymbol{u} - \boldsymbol{x}_1), \qquad c(\sigma) = 1 - t = \frac{V}{\sigma^2 + V}. \tag{2}$$

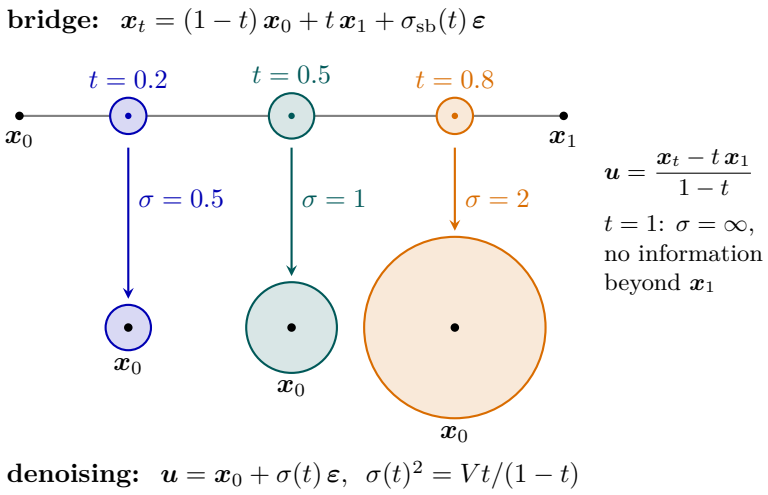


Figure 1: Every bridge time is a denoising problem. Top: bridge states at three times, with their noise $\sigma_{\mathrm{sb}}(t)$ (shaded), which vanishes at both ends. Bottom: the same states after removing $t\,\boldsymbol{x}_1$ and rescaling by $1/(1-t)$; each is the target $\boldsymbol{x}_0$ plus noise $\sigma(t)$, which grows without bound toward $\boldsymbol{x}_1$. The network sees the top row, and its optimal output $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$ is the denoiser of the bottom one. Radii drawn to scale for $V = 1$.

Training

The I^{2}SB loss $\mathbb{E}\|D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{x}_0\|^2$, with $t \sim \mathcal{U}[0,1]$, is ordinary conditional VE denoising, $D_\theta \approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$, with three design choices in EDM's [3] vocabulary:

- **input preconditioning** (2): the noisy state is shrunk toward $\boldsymbol{x}_1$ by $c(\sigma)$, so the noise the network sees, $c(\sigma)\sigma = \sigma_{\mathrm{sb}}$, never exceeds τ, while σ runs to ∞;
- **noise-level distribution**: $t \sim \mathcal{U}[0,1]$ is $p(\sigma) = 2\sigma V/(\sigma^2 + V)^2$;
- **loss**: plain $\boldsymbol{x}_0$-regression, uniform weight.

Sampling

Proposition 1. *Let $s < t$, $r = \sigma_s/\sigma_t$, and $\hat{\boldsymbol{x}}_0 = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1)$. I^{2}SB's reverse step $\boldsymbol{x}_s \sim \mathcal{N}((1 - \frac{s}{t})\hat{\boldsymbol{x}}_0 + \frac{s}{t}\boldsymbol{x}_t,\ V\frac{s(t-s)}{t}\boldsymbol{I})$ is, in the coordinates (1),*

$$\boldsymbol{u}_s = \boldsymbol{u}_t + (\sigma_t^2 - \sigma_s^2)\,\frac{\hat{\boldsymbol{x}}_0 - \boldsymbol{u}_t}{\sigma_t^2} + \sqrt{\frac{\sigma_s^2(\sigma_t^2 - \sigma_s^2)}{\sigma_t^2}}\,\boldsymbol{z}, \qquad \boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}), \tag{3}$$

the VE ancestral step, with $(\hat{\boldsymbol{x}}_0 - \boldsymbol{u}_t)/\sigma_t^2 = \nabla \log p_{\sigma_t}(\boldsymbol{u}_t \mid \boldsymbol{x}_1)$ by Tweedie's formula.

Proof. Substitute $\boldsymbol{x}_t = (1-t)\boldsymbol{u}_t + t\boldsymbol{x}_1$ into the mean and divide by $1 - s$: the $\boldsymbol{x}_1$ terms cancel, and $\mathbb{E}[\boldsymbol{u}_s] = \frac{s(1-t)}{t(1-s)}\boldsymbol{u}_t + \frac{t-s}{t(1-s)}\hat{\boldsymbol{x}}_0 = r^2\boldsymbol{u}_t + (1-r^2)\hat{\boldsymbol{x}}_0$, using $r^2 = \frac{s(1-t)}{t(1-s)}$ from (1). The variance divided by $(1-s)^2$ is $\sigma_s^2\frac{t-s}{t(1-s)} = \sigma_s^2(1-r^2)$. Both are (3). □

The bridge enters only when mapping back, $\boldsymbol{x}_s = \boldsymbol{x}_1 + c(\sigma_s)(\boldsymbol{u}_s - \boldsymbol{x}_1)$: each denoiser output is used exactly as in a VE sampler, then rescaled and offset by bridge factors.

The warm start is the first step. At $t = 1$, $\sigma = \infty$ and $r = 0$, so (3) keeps nothing of the current state:

$$\boldsymbol{u}_{t_1} = D_\theta(\boldsymbol{x}_1, 1, \boldsymbol{x}_1) + \sigma(t_1)\,\boldsymbol{z} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] + \sigma(t_1)\,\boldsymbol{z}.$$

That is: regress, then re-noise to $\sigma(t_1)$, then diffuse. This is the structure of SDEdit [4] and come-closer-diffuse-faster [6], with the initialisation supplied by the network's own one-shot regression. The last step ($s = 0$, $\sigma = 0$) returns $\hat{\boldsymbol{x}}_0$.

What the knobs become.

- τ **sets the noise scale.** On a fixed t-grid every σ_k scales with $\sqrt{V} = 2\tau$, so τ decides how warm the start is. At 20 uniform steps ($t_1 \approx 0.95$), $\sigma(t_1) \approx 1.7$ for $\tau = 0.19$ and ≈ 0.8 for $\tau = 0.09$, against image intensities ~ 1.
- **The start** $\boldsymbol{x}_1$ (zero-filled, ACS, guide) changes only the conditioning variable and the centre of the preconditioning (2).
- **Data consistency** (the CG step of I^{2}SB-Recon) replaces $\hat{\boldsymbol{x}}_0$ by an estimate of $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$ and leaves (3) unchanged (next section).

With a measurement: I^{2}SB-Recon is ImMAP2

Conditioning also on a measurement $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$, with $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_n^2 \boldsymbol{I})$, changes only the target: $\boldsymbol{u}$ is then a VE diffusion on the posterior $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1, \boldsymbol{y})$, and since $\boldsymbol{\varepsilon}$ is independent of $\boldsymbol{y}$, Tweedie still gives $\nabla \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1, \boldsymbol{y}) = (\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}] - \boldsymbol{u})/\sigma^2$. The network supplies $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$. Approximating $p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1) \approx \mathcal{N}(\hat{\boldsymbol{x}}_0, \varsigma^2 \boldsymbol{I})$ folds the data in:

$$\bar{\boldsymbol{x}}_0 = (\boldsymbol{E}^H \boldsymbol{E} + \lambda \boldsymbol{I})^{-1} (\boldsymbol{E}^H \boldsymbol{y} + \lambda \hat{\boldsymbol{x}}_0) \approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}], \qquad \lambda = \sigma_n^2/\varsigma^2, \tag{4}$$

the CG step of I^{2}SB-Recon [5]. ImMAP2 (`diffusion/immap.py`) runs, at level σ_k: denoise to $\hat{\boldsymbol{x}}_0$, apply (4), then

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}_k + h_k(\bar{\boldsymbol{x}}_0 - \boldsymbol{u}_k) + \sigma_k \sqrt{(1-\beta h_k)^2 - (1-h_k)^2}\, \boldsymbol{z}, \qquad \sigma_{k+1} = (1-\beta h_k)\, \sigma_k, \tag{5}$$

where σ_{k+1} is the level the injection aims for (ImMAP2 then re-estimates it from the next residual).

Proposition 2. *With $r = \sigma_{k+1}/\sigma_k$, $h_k = 1 - r^2$ and $\beta = 1/(1+r)$, the ImMAP2 step (5) is the I^{2}SB step (3) with $\hat{\boldsymbol{x}}_0$ replaced by $\bar{\boldsymbol{x}}_0$: I^{2}SB-Recon, written in the coordinates (1).*

Proof. $1 - \beta h_k = (1+r-1+r^2)/(1+r) = r$, so the next level is σ_{k+1}. The mean $\boldsymbol{u}_k + (1-r^2)(\bar{\boldsymbol{x}}_0 - \boldsymbol{u}_k)$ is that of (3), and the noise variance is $\sigma_k^2(r^2 - r^4) = \sigma_{k+1}^2(1-r^2)$. □

So the two methods share the step structure exactly and differ only in its settings:

	ImMAP2	I^{2}SB-Recon
denoiser	unconditional, prior $p(\boldsymbol{x}_0)$	conditional, prior $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)$
noise ladder	closed loop, $\sigma_k^2 = \frac{1}{n}\lVert\hat{\boldsymbol{x}}_0 - \boldsymbol{u}_k\rVert^2$	open loop, fixed t-grid
step size h_k	schedule $h_0 k/(1 + h_0(k-1))$	$1 - r^2$, the variance removed
β	fixed (0.05 by default)	$1/(1+r)$: 1 on the first step, $\to \frac{1}{2}$
prox weight λ	$\lambda_0 \sigma_n^2(1+\sigma_k^2)/\sigma_k^2$	constant $(7 \cdot 10^{-5})$
start	noise at $\sigma_{\max}$, or a warm start	$\boldsymbol{x}_1$ at $\sigma = \infty$: regress, re-noise

The conditional prior is the one modelling difference; the rest are schedules. ImMAP2's λ_k is (4) with $\varsigma^2 = \sigma_k^2/(1+\sigma_k^2)$, the posterior variance under a unit-variance Gaussian prior, so it follows the noise level; a constant λ assumes the denoiser is equally uncertain at every level. The start decides what the prox contributes. For the zero-filled start, $\boldsymbol{x}_1 = \boldsymbol{E}^H \boldsymbol{y}$ is sufficient for $\boldsymbol{y}$, so $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$ already uses the data and (4) only corrects the network's errors. For a guide start, (4) is the only way $\boldsymbol{y}$ enters, as in ImMAP2.

Caveats.

- The equivalence needs the *conditional* denoiser $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]$. A network that is not given $\boldsymbol{x}_1$ (no `cond_x1`, as in I^{2}SB-Inversion) learns $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t]$. It reads $\boldsymbol{x}_1$ only through $\boldsymbol{x}_t$, which works near $t = 1$ but is not the conditional denoiser in general.
- The estimation target and the sampler are the same as a VE diffusion's; the trained networks need not be. Preconditioning, $p(\sigma)$ and loss weighting are exactly the choices that shape what a finite network learns.
- For a general I^{2}SB schedule, replace $(1-t,\, t)$ by (μ_0, μ_1) and σ by σ_{sb}/μ_0; nothing else changes.

References

[1] G.-H. Liu et al. I^2SB: Image-to-Image Schrödinger Bridge. *ICML*, 2023.

[2] Y. Song et al. Score-based generative modeling through stochastic differential equations. *ICLR*, 2021.

[3] T. Karras et al. Elucidating the design space of diffusion-based generative models. *NeurIPS*, 2022.

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[6] H. Chung, B. Sim, J. C. Ye. Come-closer-diffuse-faster. *CVPR*, 2022.